15.051 Mathematical programming, HW1

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1. **Optimal production schedule**

   Setup:

Constants	
$d \in \mathbb{R}^{12}$	(product units to deliver)
$c_1 \in \mathbb{R}$	(monthly cost of storing product)
$c_2 \in \mathbb{R}$	(monthly cost of production change)
Variables	
$x \in \mathbb{R}^{12}$	(monthly production)
$\Delta x \in \mathbb{R}^{12}$	(change in monthly production)
$I \in \mathbb{R}^{12}$	(monthly inventory)
Objective	
$\sum_{i=1}^{11} c_1 I_i$	(inventory cost; none in final month)
$\sum_{i=2}^{12} c_2 \lvert\Delta x_i\rvert$	(production change cost)
Constraints	
$I_1 = 0,\ I \geq 0$	(no starting inventory; never negative inventory)
$x_1 \geq d_1,\ I_i + x_i \geq d_i$	(always meet demand)
$\Delta x_i = x_i - x_{i-1},\ i = 2, \ldots, 12$	(define change in production)
$I_i = I_{i-1} + x_i - d_i,\ i = 2, \ldots, 12$	(define inventory)

Initial formulation (not LP):

$$\begin{aligned} \min_{I,x,\Delta x} \quad & \sum_{i=1}^{11} c_1 I_i + \sum_{i=2}^{12} c_2 |\Delta x_i| \\ \text{s.t.} \quad & I_1 = 0,\ I \geq 0 \\ & x_1 \geq d_1,\ I_i + x_i \geq d_i \\ & \Delta x_i = x_i - x_{i-1},\ i = 2, \ldots, 12 \\ & I_i = I_{i-1} + x_i - d_i,\ i = 2, \ldots, 12 \end{aligned}$$

Reformulated as an LP:

$$\begin{aligned} \min_{I,x,\Delta x^+,\Delta x^-} \quad & \sum_{i=1}^{11} c_1 I_i + \sum_{i=2}^{12} c_2 (\Delta x_i^+ + \Delta x_i^-) \\ \text{s.t.} \quad & I_1 = 0,\ I \geq 0 \\ & x_1 \geq d_1,\ I_i + x_i \geq d_i \\ & \Delta x_i^+ - \Delta x_i^- = x_i - x_{i-1},\ i = 2, \ldots, 12 \\ & I_i = I_{i-1} + x_i - d_i,\ i = 2, \ldots, 12 \\ & \Delta x^+ \geq 0,\ \Delta x^- \geq 0 \end{aligned}$$

Solution:

```python
# %%
import numpy as np
import cvxpy as cp
import polars as pl

pl.Config.set_tbl_rows(100)
# %%
# TEXT PROBLEM 1
# Constants
d = np.array([5, 1, 3, 1, 3, 8, 9, 8, 9, 2, 5, 3]) * 1.0
c_1 = 4
c_2 = 10

# Variables
x = cp.Variable(len(d))
delta_x_pos = cp.Variable(len(d))
delta_x_neg = cp.Variable(len(d))
I = cp.Variable(len(d))

# Constraints
constraints = [
    I[0] == 0,  # no starting inventory
    I[1:] == I[:-1] + x[1:] - d[1:],  # inventory balance
    x[0] >= d[0],  # meet first month demand
    I >= 0,  # no negative inventory
    x >= 0,  # no negative orders
    delta_x_pos >= 0,  # no negative pos changes
    delta_x_neg >= 0,  # no negative neg changes
    (delta_x_pos - delta_x_neg)[1:] == x[1:] - x[0:-1],  # link changes to orders
]

# Objective
obj_fn = c_1 * cp.sum(I[:-1]) + c_2 * cp.sum(delta_x_pos + delta_x_neg)
obj = cp.Minimize(obj_fn)

# Solve
prob = cp.Problem(obj, constraints)
prob.solve()
```

```
✓ # TEXT PROBLEM 1 ···

Total cost:      163.0
Results:
shape: (12, 5)
```

month	demand	production	delta_prod	inventory
i64	f64	f64	f64	f64
1	5.0	5.0	0.0	0.0
2	1.0	2.0	-3.0	1.0
3	3.0	2.0	-0.0	0.0
4	1.0	2.0	-0.0	1.0
5	3.0	2.0	0.0	0.0
6	8.0	8.5	6.5	0.5
7	9.0	8.5	0.0	-0.0
8	8.0	8.5	0.0	0.5
9	9.0	8.5	-0.0	-0.0
10	2.0	3.5	-5.0	1.5
11	5.0	3.5	-0.0	0.0
12	3.0	3.5	-0.0	0.5

```
Press ⇧ Enter to execute.
```

2. **1.7 Moment problem**

Given: $Z \sim \mathcal{D}$ with $z \in \{0, 1, \ldots, K\},\ p(Z = z) = p_z,\ E[Z],\ E[Z^2]$

Let $z^n = [1, 2^n, \ldots, K^n]^T \in \mathbb{R}^K,\ p = [p_1, \ldots, p_K]^T \in \mathbb{R}^K$
Then $E[Z] = p^T z,\ E[Z^2] = p^T z^2$, and $E[Z^4] = p^T z^4$

Assume $z^n,\ K,\ E[Z],\ E[Z^2]$ are known. If p is known, $E[Z^4]$ can be trivially solved for. Assuming p is unknown, we can use an LP to maximize and minimize $E[Z^4]$ over p. These solutions provide the upper and lower bounds respectively:

$$\begin{aligned} \min_p \,/\, \max_p \quad & p^T z^4 \\ \text{s.t.} \quad & p^T z = E[Z] \\ & p^T z^2 = E[Z^2] \\ & p \geq 0,\ \mathbf{1}^T p = 1 \end{aligned}$$

3. **1.12 Chebychev center**

$$\mathcal{P} = \{x \in \mathbb{R}^n \mid a_i^T x \leq b_i,\ i = 1, \ldots, m\} \qquad \text{(polyhedron)}$$
$$\mathcal{B} = \{x \in \mathbb{R}^n \mid ||x - y||_2 \leq r\} \qquad \text{(ball with center at } y \text{ and radius } r\text{)}$$

We can use an optimization problem to solve for the ball centered at y with maximum radius, r contained within $\mathcal{P}$, but this is not an LP:

$$\begin{aligned} \max_{x,r} \quad & r \\ \text{s.t.} \quad & a_i^T x \leq b_i,\ i = 1, \ldots, m \\ & ||x - y||_2 \leq r \\ & r \geq 0 \end{aligned}$$

We can substitute nonlinear constraint $||x - y||_2 \leq r$ with a requirement that every line segment from ball center, y, to every hyperplane, $a_i^T x = b_i$, is less than or equal to r. By definition, distance, d, from y to a point, x_{i_0}, on hyperplane i is

$$d = \frac{a_i}{||a_i||_2}^T (x_{i_0} - y) = \frac{a_i^T x_{i_0} - a_i^T y}{||a_i||_2} = \frac{b_i - a_i^T y}{||a_i||_2}$$

Note on sign: distance is positive in this equation when $a_i^T y \leq b_i$, which is directly what we want for $\mathcal{B} \subseteq \mathcal{P}$
So an LP that maximizes ball radius r subject to $\mathcal{B} \subseteq \mathcal{P}$ can be formulated as

$$\begin{aligned} \max \quad & r \\ \text{s.t.} \quad & r \leq \frac{b_i - a_i^T y}{||a_i||_2},\ i = 1, \ldots, m \\ & r \geq 0 \end{aligned}$$

4. **2.6 Caratheodory theorem**

Given: $A_1, \ldots, A_n \in \mathbb{R}^m$

(a) let $\mathcal{C} = \{\sum_i^n \lambda_i A_i \mid \lambda_i \geq 0\}$ and $A = [A_1, \ldots, A_n]$
Show: For any $x = A\lambda \in \mathcal{C}$, $\exists \lambda^* \geq 0$ with $|supp(\lambda*)| \leq m$ and $x = A\lambda^*$

Trivial case ($n \leq m$): by definition we have $\lambda^* = \lambda$, since any $x \in \mathcal{C} = A\lambda$ has $|supp(\lambda)| \leq n \leq m$

Main case ($n > m$): Approach is to show that an algorithm exists to construct λ^*.

Algorithm: First, choose any $x \in \mathcal{C} = A\lambda = \sum_{i=1}^n \lambda_i a_i$

1. If $|supp(\lambda)| \leq m$ then stop. We have $\lambda^* = \lambda$
2. Take any new nonzero vector $d \in \mathcal{N}(A)$ with $supp(d) \subseteq supp(\lambda)$.
   We know d exists since since $supp(\lambda) > m$ and $\mathcal{R}(A) \leq m$ so the columns of A that are supported by $supp(\lambda)$ are linearly dependent.
   Since $Ad = 0$ by definition, we can rewrite $A\lambda = A\lambda - \alpha Ad = A(\lambda - \alpha d)$
3. Choose α such that $\lambda' = \lambda - \alpha d \geq 0$. This is possible since we can let

$$\alpha = min_i \frac{\lambda_i}{d_i}, \text{ over all } i \text{ where } d_i > 0$$

   If $d \leq 0$, we can define new $d := -d$. Now by construction, we have zeroed out index i_{min} since $\lambda'_{i_{min}} = \lambda_{i_{min}} - \frac{\lambda_{i_{min}}}{d_{i_{min}}} d_{i_{min}} = 0$. Feasibility check on the remaining indices:
   - $d_i > 0$: $\lambda'_j = \lambda_j - \frac{\lambda_{i_{min}}}{d_{i_{min}}} * d_j \Longrightarrow \frac{\lambda'_j}{d_j} = \frac{\lambda_j}{d_j} - \frac{\lambda_{i_{min}}}{d_{i_{min}}} \geq 0$ since $\frac{\lambda_j}{d_j} \geq \frac{\lambda_{i_{min}}}{d_{i_{min}}}$
   - $d_i = 0$: $\lambda'_i = \lambda_i - \alpha 0 = \lambda_i \geq 0$
   - $d_i < 0$: $\lambda'_i \geq \lambda_i \geq 0$ since $\alpha \geq 0$
4. Now we have constructed λ' with $|supp(\lambda')| < |supp(\lambda)|$. Return to step 1 and use λ' as λ

(b) let $\mathcal{P} = \{A\lambda \mid \lambda \geq 0, \mathbf{1}^T\lambda = 1\}$
Show: For any $x = A\lambda \in \mathcal{P}$, $\exists \lambda^* \geq 0$ with $\mathbf{1}^T\lambda^* = 1$, $|supp(\lambda^*)| \leq m + 1$, and $x = A\lambda^*$

Trivial case ($n < m$): by definition we have $\lambda^* = \lambda$, since any $x \in \mathcal{P} = A\lambda$ has $|supp(\lambda)| \leq n < m < m + 1$

Main case ($n \geq m$): The approach is to reframe the constraints in set $\mathcal{P}$ so that we can call on part (a). By definition, for any $x \in \mathcal{P}$, we have the following system of equations:

$$\begin{bmatrix} A \\ \mathbf{1}^T \end{bmatrix} \lambda = \begin{bmatrix} x \\ 1 \end{bmatrix}$$

We can define $\hat{A} := \begin{bmatrix} A \\ \mathbf{1}^T \end{bmatrix} \in \mathbb{R}^{m+1 \times n}$ and repeat the algorithm in part (a) using $\hat{A}$ in place of A and $\mathcal{R}(\hat{A}) \leq m+1$.

This demonstrates that we can construct λ^* with $|supp(\lambda^*)| \leq m+1$. And we know $\hat{A}\lambda^* = \begin{bmatrix} A \\ \mathbf{1}^T \end{bmatrix} \lambda^* = \begin{bmatrix} x \\ 1 \end{bmatrix}$ and $\lambda^* \geq 0$ by construction.

5. **2.13 Basic feasible solution**
Given: Standard form polyhedron $\mathcal{P} = \{x \mid Ax = b, x \geq 0\}$, $A \in \mathbb{R}^{m \times n}$ has m linearly independent rows, and all basic feasible solutions are nondegenerate.

Let $x \in \mathcal{P}$ with exactly m positive components

(a) **Show:** x is basic feasible solution
Let $I = \{i \mid x_i \neq 0\}$ and $N = I^C$.

Contradiction: Suppose x is not a basic feasible solution. Since x is not a BFS, but has m nonzero values, then the only way it fails BT Theorem 2.4 is that the columns of $A_I = [A_i \text{ for } i \in I] \in \mathbb{R}^{m \times m}$ are linearly dependent (if they were independent, there would be a unique solution to $A_I x_I = b$ and x would be a BFS).

Then there exists a nonzero $d_I \in \mathbb{R}^m$ such that $A_I d_I = 0$. For $x \in \mathcal{P}$, let $y = x - \alpha d$. If we choose $\alpha = \min_{i \in I | d_i > 0} x_i / d_i$ we have that

- $x_i - \alpha d_i \geq 0 \quad \forall i \in I \Rightarrow y_I \geq 0$
- $A_I y_I = A_I(x_I - \alpha d_I) = A_I x_I - \alpha A_I d_I = A_I x_I + 0 = b$
- $y_{i_{min}} = x_{i_{min}} - \frac{x_{i_{min}}}{d_{i_{min}}} d_{i_{min}} = 0$

And hence $|supp(y_I)| < m$. Note, for the remaining $j \in N$, we can trivially set $d_j = 0 \Rightarrow y_j = 0$ and see that $y \in \mathcal{P}$ and hence $|supp(y)| < m$.

We may continue this dimension elimination process until we have produced y^* with $|supp(y^*)| = l < m$ and linearly independent columns A_l for $l \in L := \{l | y_l^* \neq 0\}$. We have constructed a BFS, y^*, with $l < m$ positive components, which by definition is a degenerate BFS and contradicts the given that all basic feasible solutions are nondegenerate.

(b) **Show:** The result in part (a) fails if the nondegeneracy assumption is removed.

This follows directly from our proof in part (a), where we can no longer build a contradiction to the given assumption that all basic feasible solutions are nondegenerate. To emphasize this point, I provide a counterexample. Let

$$\text{Let } A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \in \mathbb{R}^{2 \times 3} \text{ and } b = [2, 0]^T$$

Defining standard form $\mathcal{P} = \{x \in \mathbb{R}^3 \mid Ax = b, x \geq 0\}$, we can choose $x = [1, 1, 0]^T \in \mathcal{P}$ and we have set up an example of the stated problem:

- A has $m = 2$ linearly independent rows
- x has exactly $m = 2$ positive components

Letting $I = \{i \mid x_i \neq 0\}$, we have $A_I = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$, which does not have linearly independent columns and hence $A_I x_I = b$ does not have a unique solution and hence x is not a basic feasible solution.